Mat 1575 HW9 (§3.4)

Express the rational function as a sum or difference of two simpler rational expressions.

183. $\dfrac{x^2+1}{x(x+1)(x+2)}$ (deg 2 / deg 3) ⇒ Can do Partial Fraction Decomposition Directly.

$$= \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2} \Rightarrow \frac{x^2+1}{x(x+1)(x+2)} = \frac{A(x+1)(x+2)+Bx(x+2)+Cx(x+1)}{x(x+1)(x+2)}$$

$$\Rightarrow x^2+1 = A(x+1)(x+2)+B(x)(x+2)+C(x)(x+1)$$

- To find A, let $x=0$, we have $0^2+1 = A(0+1)(0+2)+B(0)(0+2)+C(0)(0+1)$

$$\Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$$

- To find B, let $x=-1$, we have $(-1)^2+1 = A(-1+1)(-1+2)+B(-1)(-1+2)+C(-1)(-1+1)$

$$\Rightarrow 2 = -B, \quad B = -2$$

- To find C, let $x=-2$, we have $(-2)^2+1 = A(-2+1)(-2+2)+B(-2)(-2+2)+C(-2)(-2+1)$

$$\Rightarrow 5 = 2C, \quad C = \frac{5}{2}$$

$$\Rightarrow \frac{x^2+1}{x(x+1)(x+2)} = \frac{\frac{1}{2}}{x} + \frac{-2}{x+1} + \frac{\frac{5}{2}}{x+2}$$

185. $\dfrac{3x+1}{x^2}$ (deg 1 / deg 2) ⇒ Can do Partial Fraction Decomposition Directly. But there is an easy way ....

$$= \frac{3x}{x^2} + \frac{1}{x^2} = \frac{3}{x} + \frac{1}{x^2}.$$

187. $\dfrac{2x^4}{x^2-2x}$ (deg 4 / deg 2) ⇒ Simplify it first (or do long division ...)

$$= \frac{2x^4-4x^3+4x^3-8x^2+8x^2-16x+16x}{x^2-2x} = \frac{2x^2(x^2-2x)+4x(x^2-2x)+8(x^2-2x)+16x}{x^2-2x}$$

$$= \frac{2x^2(x^2-2x)}{x^2-2x} + \frac{4x(x^2-2x)}{x^2-2x} + \frac{8(x^2-2x)}{x^2-2x} + \frac{16x}{x^2-2x} = 2x^2+4x+8+\frac{16x}{x^2-2x}$$

$$= 2x^2+4x+8+\frac{16x}{x(x-2)} = 2x^2+4x+8+\frac{16}{x-2} \quad (\text{if } x \neq 0)$$

Use the method of partial fractions to evaluate each of the following integrals.

196. $\int \frac{dx}{(x-3)(x-2)} = \int \left(\frac{A}{x-3} + \frac{B}{x-2}\right)dx$

$= \int \left(\frac{1}{x-3} + \frac{-1}{x-2}\right)dx$

$= \ln|x-3| - \ln|x-2| + C$

$\frac{1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2} = \frac{A(x-2)+B(x-3)}{(x-3)(x-2)}$

$\Rightarrow 1 = A(x-2)+B(x-3)$

- $x=3 \Rightarrow 1 = A$.
- $x=2 \Rightarrow 1 = -B \Rightarrow B=-1$

197. $\int \frac{3x}{x^2+2x-8}dx = \int \frac{3x}{(x+4)(x-2)}dx$

$= \int \left(\frac{2}{x+4} + \frac{1}{x-2}\right)dx$

$= 2\ln|x+4| + \ln|x-2| + C$

$\frac{3x}{(x+4)(x-2)} = \frac{A}{(x+4)} + \frac{B}{(x-2)} = \frac{A(x-2)+B(x+4)}{(x-2)(x+4)}$

$\Rightarrow 3x = A(x-2)+B(x+4)$

- $x=-4 \Rightarrow -12 = -6A \Rightarrow A=2$
- $x=2 \Rightarrow 6 = 6B \Rightarrow B=1$

199. $\int \frac{x}{x^2-4}dx = \int \frac{x}{(x+2)(x-2)}dx$

$= \int \left(\frac{\frac{1}{2}}{x+2} + \frac{\frac{1}{2}}{x-2}\right)dx$

$= \frac{1}{2}\ln|x+2| + \frac{1}{2}\ln|x-2| + C$

$\frac{x}{(x+2)(x-2)} = \frac{A}{x+2} + \frac{B}{x-2} = \frac{A(x-2)+B(x+2)}{(x+2)(x-2)}$

$\Rightarrow x = A(x-2)+B(x+2)$

- $x=-2 \Rightarrow -2 = A(-4) \Rightarrow A=\frac{1}{2}$
- $x=2 \Rightarrow 2 = 4B \Rightarrow B=\frac{1}{2}$

200. $\int \frac{dx}{x(x-1)(x-2)(x-3)}$

$= \int \left(\frac{-\frac{1}{6}}{x} + \frac{\frac{1}{2}}{x-1} + \frac{-\frac{1}{2}}{x-2} + \frac{\frac{1}{6}}{x-3}\right)dx$

$= -\frac{1}{6}\ln|x| + \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x-2| + \frac{1}{6}\ln|x-3| + C$

$\frac{1}{x(x-1)(x-2)(x-3)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x-2} + \frac{D}{x-3}$

$1 = A(x-1)(x-2)(x-3) + Bx(x-2)(x-3) + Cx(x-1)(x-3) + Dx(x-1)(x-2)$

- $x=0 \Rightarrow 1 = -6A \Rightarrow A=-\frac{1}{6}$
- $x=1 \Rightarrow 1 = 2B \Rightarrow B=\frac{1}{2}$
- $x=2 \Rightarrow 1 = -2C \Rightarrow C=-\frac{1}{2}$
- $x=3 \Rightarrow 1 = 6D \Rightarrow D=\frac{1}{6}$

201. $\int \frac{2x^2+4x+22}{x^2+2x+10}\,dx$ deg 2 / deg 2 — Simplify it first then do decomposition later

$= \int \frac{2x^2+4x+20+2}{x^2+2x+10}\,dx = \int \frac{2x^2+4x+20}{x^2+2x+10} + \frac{2}{x^2+2x+10}\,dx$

$= \int \frac{2(x^2+2x+10)}{x^2+2x+10} + \frac{2}{x^2+2x+10}\,dx = \int 2 + \frac{2}{x^2+2x+10}\,dx$ — $x^2+2x+10$ can't be further factorized

Here we do "complete the square" $= \int 2 + \frac{2}{(x+1)^2+9}\,dx = \int 2 + \frac{\frac{2}{9}}{\frac{(x+1)^2}{9}+\frac{9}{9}}\,dx = \int 2 + \frac{\frac{2}{9}}{(\frac{x+1}{3})^2+1}\,dx$

to reveal the "$\tan^{-1}(x)$" $= 2x + \frac{2}{9}\cdot 3\tan^{-1}\left(\frac{x+1}{3}\right) + C = 2x + \frac{2}{3}\tan^{-1}\left(\frac{x+1}{3}\right) + C$

$\int \frac{dx}{x^2+1} = \tan^{-1}(x) + C$

202. $\int \frac{dx}{x^2-5x+6} = \int \frac{dx}{(x-2)(x-3)} = \int \frac{-1}{x-2} + \frac{+1}{x-3}\,dx$

$= -\ln|x-2| + \ln|x-3| + C$

203. $\int \frac{2-x}{x^2+x}\,dx = \int \frac{2-x}{x(x+1)}\,dx = \int \frac{2}{x} + \frac{-3}{x+1}\,dx$

$= 2\ln|x| - 3\ln|x+1| + C$

204. $\int \frac{2}{x^2-x-6}\,dx = \int \frac{2}{(x+2)(x-3)}\,dx = \int \frac{\frac{2}{-5}}{x+2} + \frac{\frac{2}{5}}{x-3}\,dx$

$= -\frac{2}{5}\ln|x+2| + \frac{2}{5}\ln|x-3| + C$